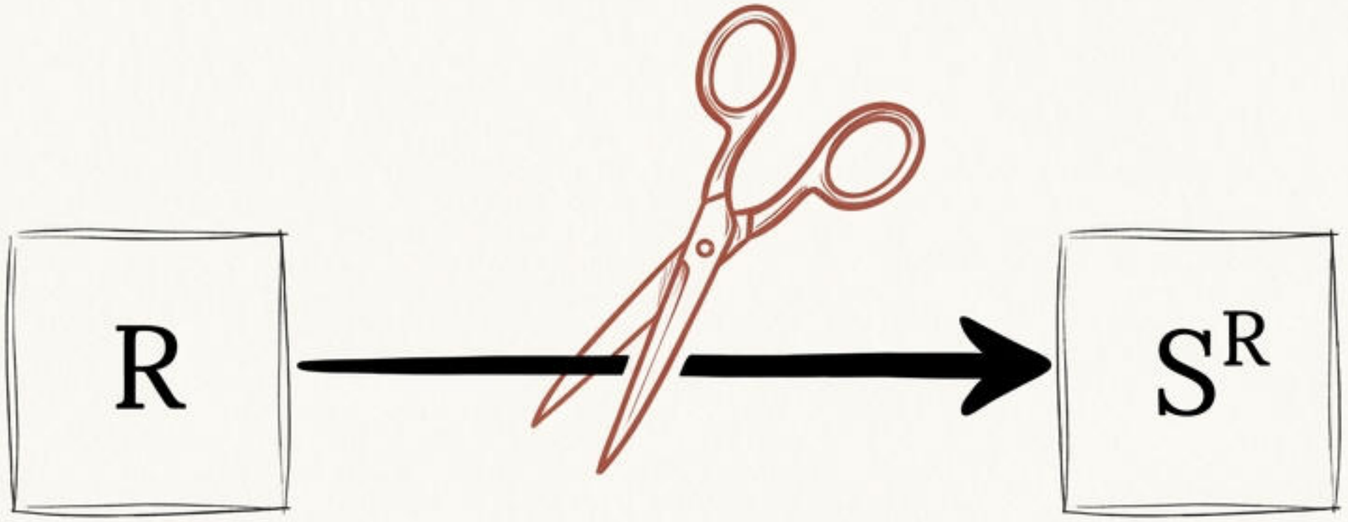
Operant Extinction: Processes and Resistance

A Deep Dive into the Weakening of **Operant Behavior**



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Defining the Extinction Process

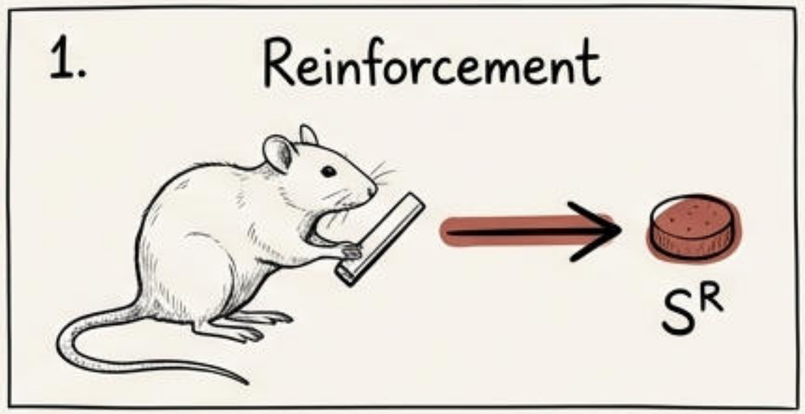
Operant extinction is a fundamental process for weakening behavior, but it is crucial to distinguish between its two components: the **procedure** and the **process**.

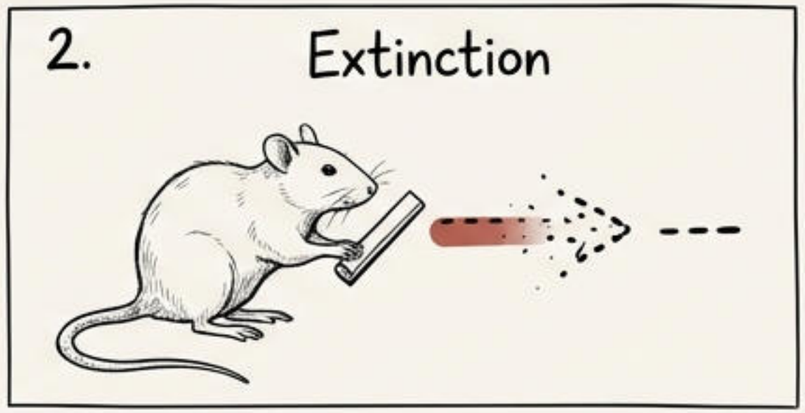
- **The Procedure of Extinction:** This refers to the specific action of *nonreinforcement* of a previously reinforced response. It is the environmental arrangement where the reinforcing consequence is actively withheld or no longer delivered following the behavior.
- **The Process of Extinction:** This is the *resultant decrease* in the strength of the response. The outcome of the procedure is a gradual decline in the frequency, intensity, or duration of the behavior.

This relationship can be represented formally. While reinforcement establishes a contingency, extinction breaks it:

- **Reinforcement:** Lever Press → Food Pellet
$$R \rightarrow S^R$$
- **Extinction:** Lever Press → No Food Pellet
$$R \rightarrow ---$$

If the frequency of lever pressing declines as a result of withholding food, we are witnessing the process of extinction. If responding ceases entirely, the response is **extinguished**. If it occurs at a lower rate, it is **partially extinguished**.





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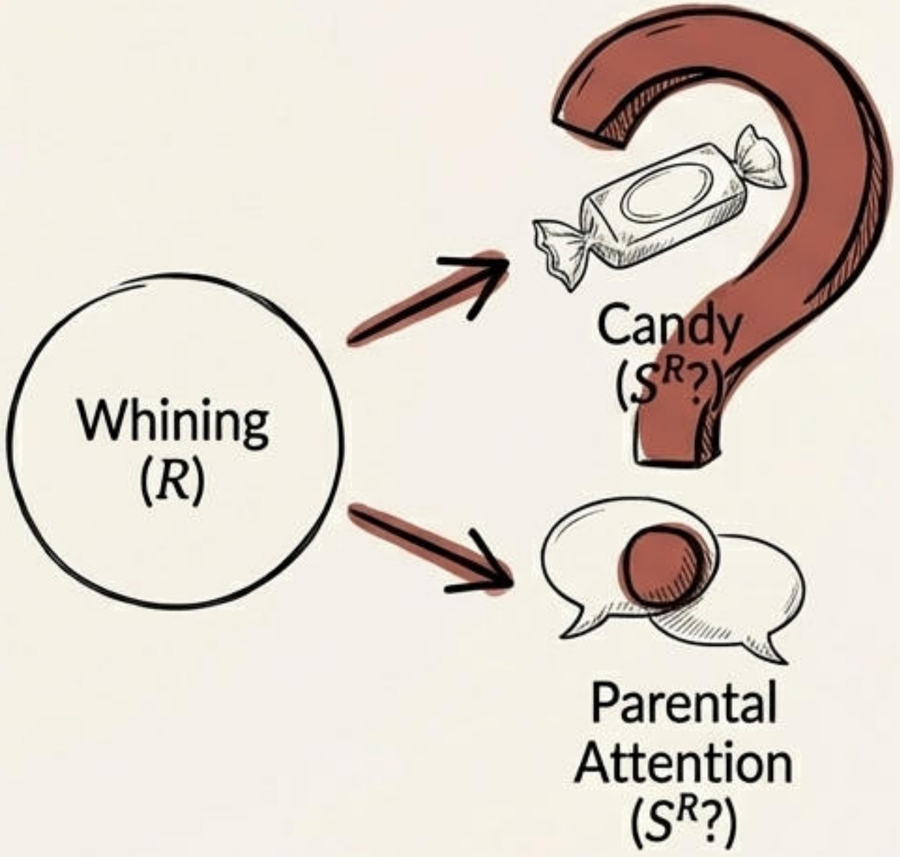
Identifying the Effective Reinforcer

A critical, and often overlooked, step in applying an extinction procedure is ensuring that the consequence being withheld is, in fact, the *actual reinforcer* maintaining the behavior. A failure in this functional analysis will render the procedure ineffective.

Consider a child who whines in a supermarket:

- **Apparent Contingency:** Whining → Candy ($R \rightarrow S^R$)
- **Extinction Procedure:** Whining → No Candy ($R \rightarrow$ ---)

However, what if the candy is merely incidental and the true reinforcer is the parental attention (e.g., arguing, scolding, explaining) that accompanies the whining? If the parent withholds the candy but continues to provide attention each time the child whines, the extinction procedure for candy will have little effect. **It is even possible that both consequences—candy and attention—are functioning as** as reinforcers. In this case, withholding only the candy would, at best, only partially extinguish the whining.



Therefore, determining the effective reinforcer(s) maintaining a behavior is the foundational first step before any extinction procedure can be successfully designed and implemented.

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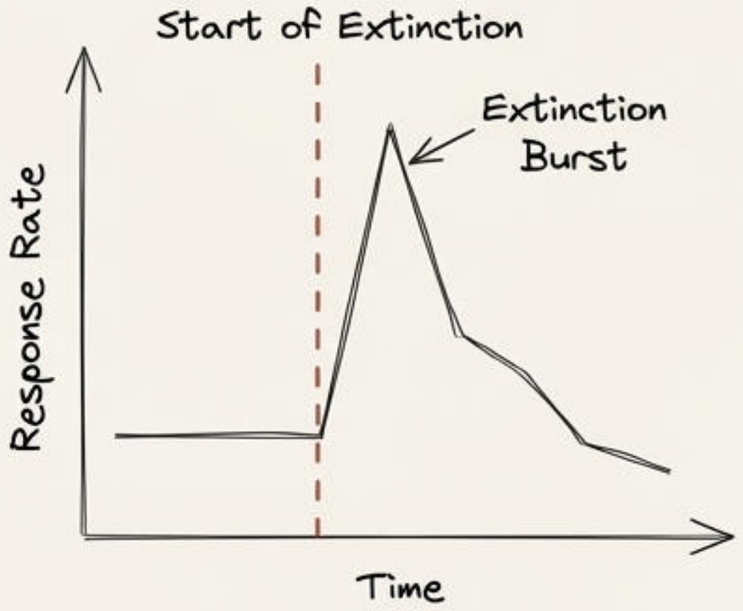
Behavioral Side Effects of Extinction

Implementing an extinction procedure rarely results in a smooth, immediate decline in behavior. The process is often accompanied by several predictable side effects. Understanding these effects is crucial, as they can be misinterpreted as the procedure failing when it is actually beginning to work.

1. Extinction Burst

This is a temporary increase in the **frequency and intensity** of responding when extinction is first implemented. The organism does not immediately cease responding; instead, it initially responds with greater vigor.

- **Laboratory Example:** A rat reinforced on an FR 4 schedule for lever pressing, when placed on extinction, will **not just stop. It will often begin pressing the lever more rapidly and with greater force.**
- **Human Analogy:** The Uncooperative Vending Machine. If you put money in a vending machine and press the button, but nothing comes out, you don't immediately walk away. Your first reaction is to press the button again, several times, perhaps faster and harder. **This increase in frequency and force is a classic extinction burst.**



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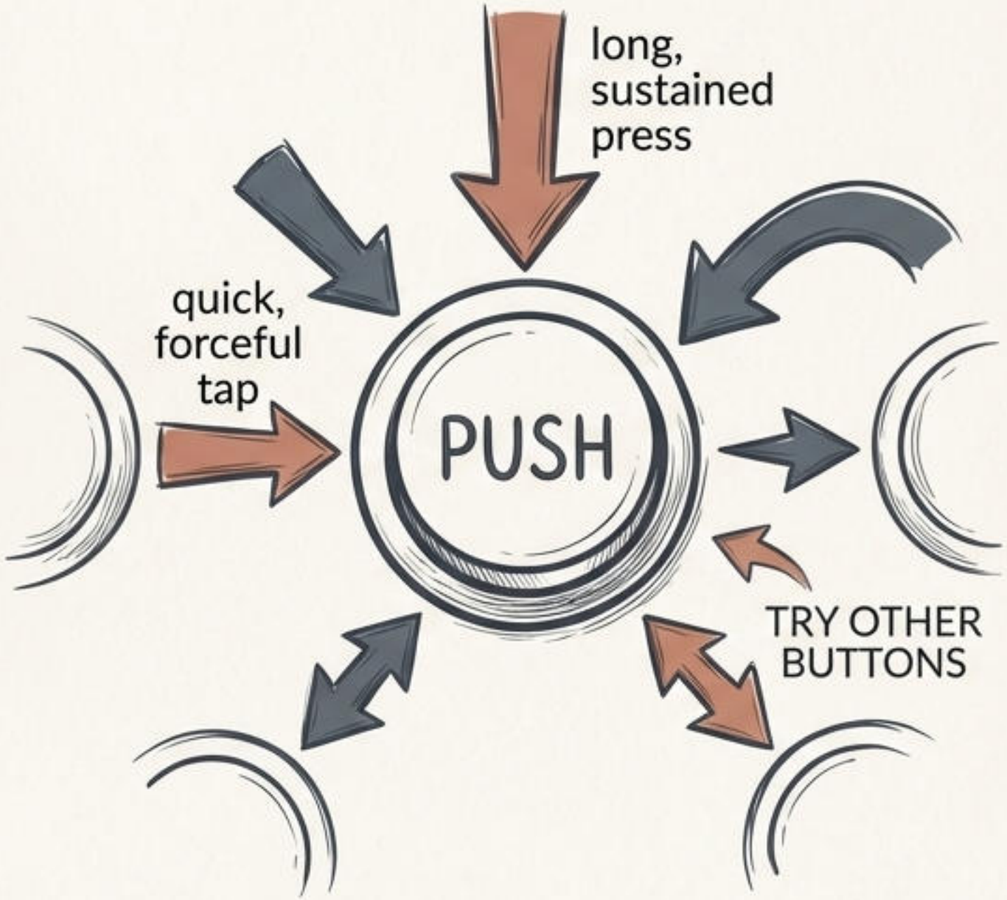
2. Increase in Variability

An extinction procedure often produces an increase in the **variability of the behavior**. The topography of the response begins to change as the organism tries other ways to elicit the reinforcer.

- **Laboratory Example:** A rat that typically pressed a lever with its right paw might begin pressing it with its left paw, its nose, or on a different part of the lever. It is trying variants of the original successful response.
- **Human Analogy: The Vending Machine Revisited.** When repeated, forceful button pressing fails, you increase the variability of your behavior. You might try holding the button down for a longer duration. You will almost certainly try pressing other buttons for different selections, hoping *something* will be delivered.



Note: Some theorists consider an increase in response variability to be another dimension of the extinction burst itself—an increase in the rate, intensity, and variability of responding following the implementation of an extinction procedure.

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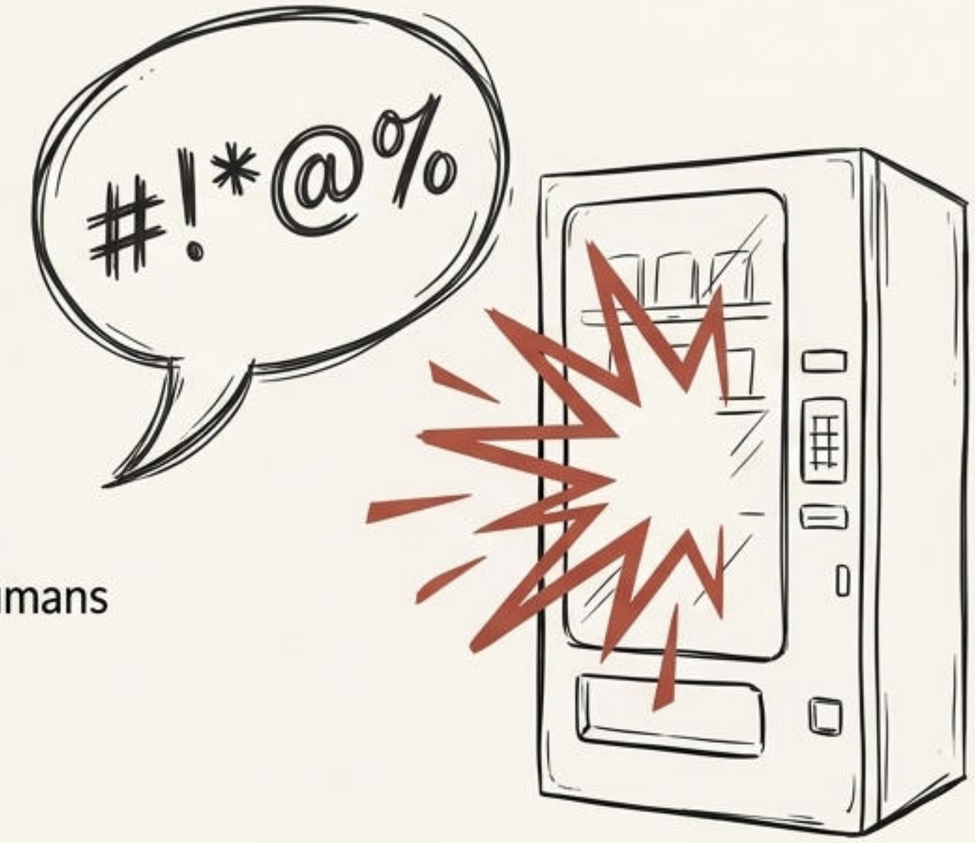
3. Emotional Behavior & Aggression

Extinction is frequently accompanied by emotional responses, which are often labeled as **frustration**. This can escalate into overt aggression, a phenomenon known as **extinction-induced aggression** (or frustration-induced aggression).

- **Emotional Behavior:** A hungry pigeon that finds its key-pecking no longer produces food will become agitated, evidenced by jerky movements and wing flapping. Humans confronted by a malfunctioning machine often become visibly upset.
- **Aggression:** This frustration can lead to direct aggressive acts. In classic studies (Azrin, Hutchinson, & Hake, 1966), a pigeon whose key-pecking is put on extinction will reliably attack another pigeon (or even an inanimate model of a pigeon) placed in the chamber. Humans are no different; uncope vending machines are sometimes kicked, shaken, or struck. This links directly to real-world phenomena like "**road rage**" or "**air rage**," which can be functionally analyzed as adult-scale temper tantrums—aggressive outbursts in response to the frustration of a blocked goal (i.e., extinction).



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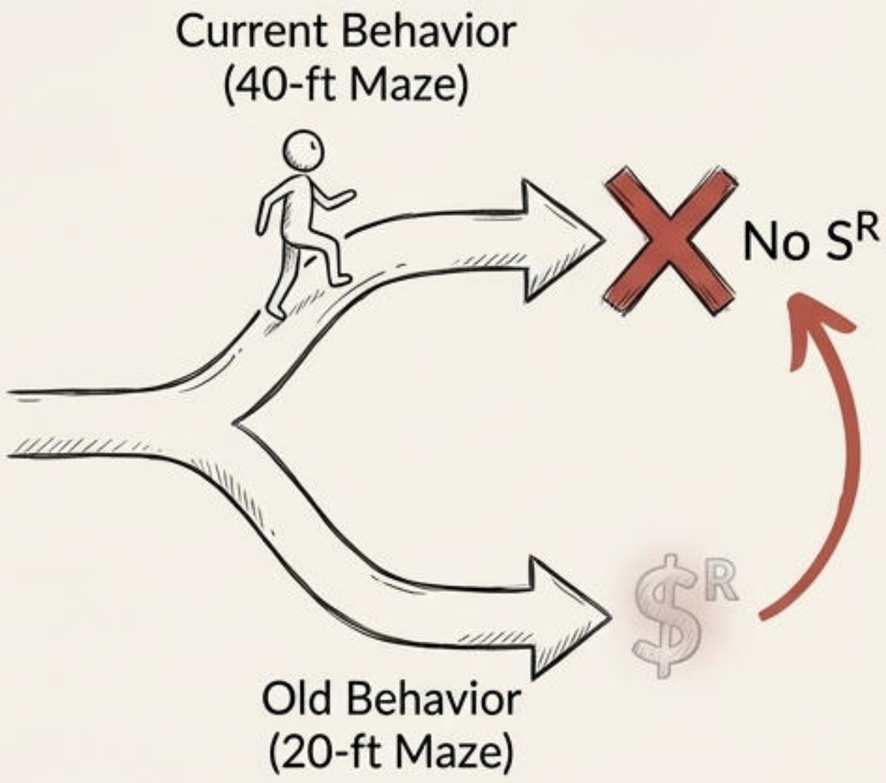
4. Resurgence

Resurgence is the reappearance during extinction of **other behaviors that had once been effective** in obtaining reinforcement in the past. When the currently trained behavior is no longer reinforced, the organism may revert to an older, previously successful strategy.

- **Laboratory Example (Hull, 1934):** Rats were first trained to run a 20-foot maze for food, then a longer 40-foot maze. When reinforcement was stopped entirely (extinction), the rats initially persisted with the 40-foot pattern. When that failed, they **reverted to running the old 20-foot pattern before eventually quitting.** They **returned to a behavior that had previously worked.**

- **Human Analogy: The Stressed Husband.** A husband feeling that his marriage is providing very little reinforcement (his bids for connection are on extinction) might start spending **increasing amounts of time at his parents' house.** He is returning to a setting that was, in the past, a rich and reliable source of reinforcement.

This behavioral process closely resembles the psychoanalytic concept of **regression**, the reappearance of immature behavior in reaction to frustration or conflict.



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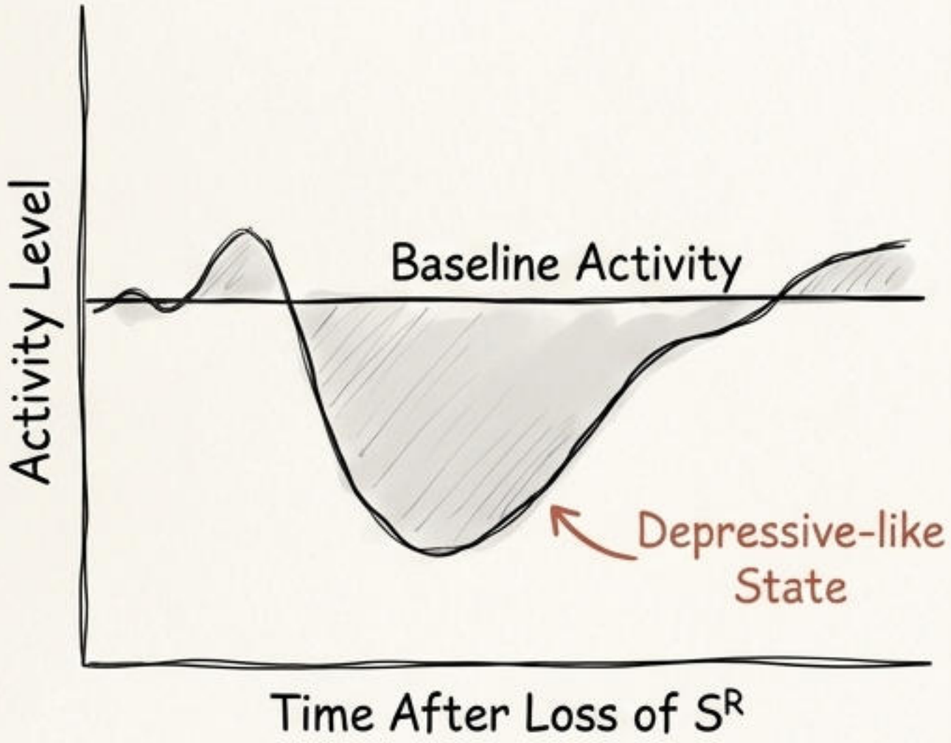
5. Depression

Following the **loss of a major source of reinforcement**, **extinction** can lead to **a period of depressive-like symptoms**, most notably a **decrease in general activity level.**

- Laboratory Example (Klinger et al., 1974): Rats were trained to **run down an alley** for food, after which their activity was measured in an open field. When the food was removed (extinction), their subsequent activity in the open field first showed a brief increase, but then **dropped to below-normal levels**, before eventually returning to baseline.



This finding has profound parallels with human **depression,** which is often **clinically associated with a major loss of reinforcement (e.g., the death of a loved one, loss of a job).** The people for whom that person or activity was a major source of reinforcement are effectively on an extinction schedule. A key symptom of this state is a low level of activity (anhedonia, lethargy). The animal data suggests that a **temporary period of depression,** accompanied by **decreased activity**, may be a normal, adaptive aspect of disengaging from a major, now-unavailable reinforcer.

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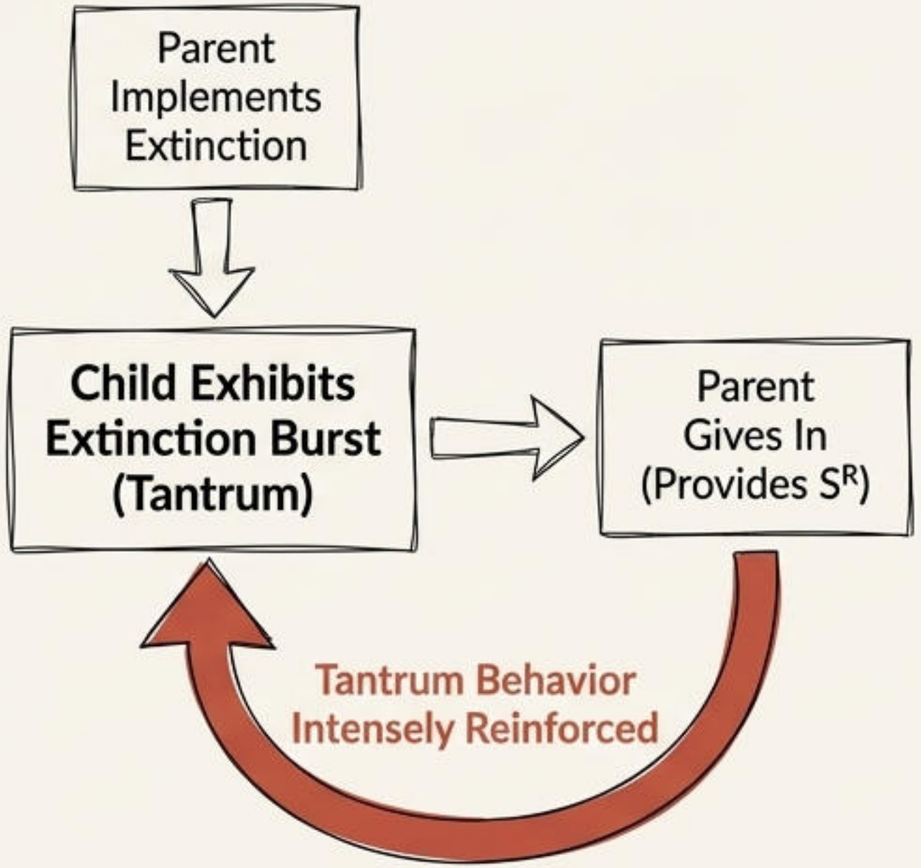
Clinical Implications: The Danger of Giving In

The side effects of extinction carry significant clinical implications. If an individual gives in during a side effect, they risk powerfully reinforcing the very behavior they are trying to eliminate—often in its most extreme form. Consider a parent attempting to extinguish a child's begging for candy at the supermarket:

1. **Procedure:** Parent withholds candy.
2. **Side Effects:** The child, Bobbie, exhibits an **extinction burst** (louder, more persistent begging), **emotional behavior** (crying), and **aggression** (kicking, throwing a tantrum).
3. **Reinforcement:** Overwhelmed by the tantrum, the parent gives in and buys the candy.

In this scenario, the parent has not reinforced polite asking. They have delivered a powerful reinforcer contingent upon a severe temper tantrum. This inadvertently shapes more intense and problematic behavior in the future. Repeated cycles of this interaction can establish deeply ingrained patterns of tantrums that may persist into adulthood, manifesting as phenomena like "road rage." The behavior pattern was established and strengthened through intermittent reinforcement of its most intense variations.

The Tantrum Cycle



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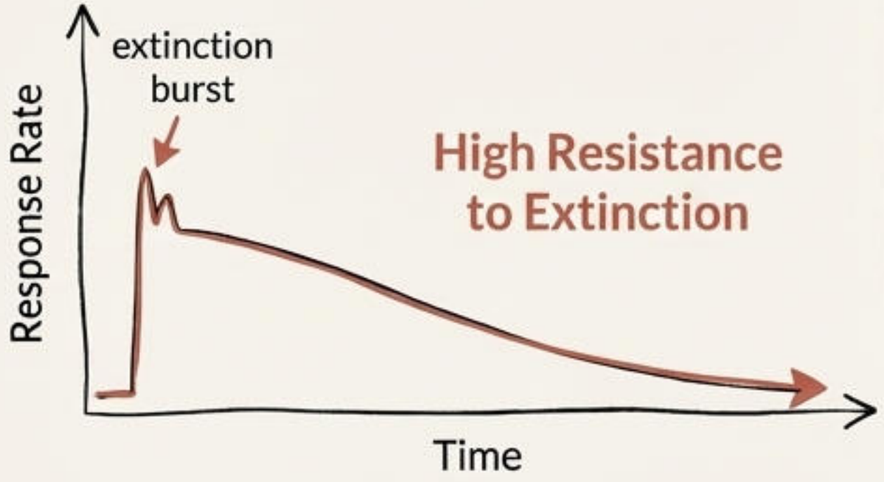
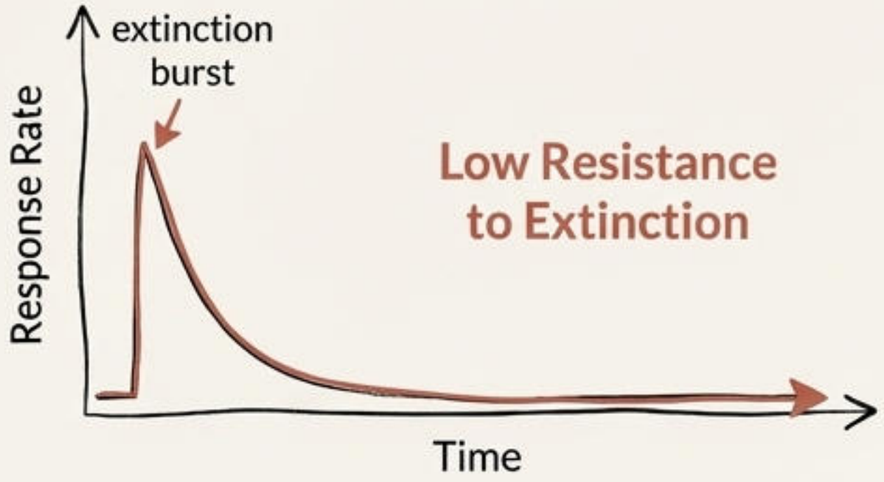
Factors Influencing Resistance to Extinction

Resistance to extinction is the extent to which responding persists after an extinction procedure has been implemented. A response that continues for a long time has **high resistance**, while a response that disappears quickly has **low resistance**.

For example, a dog that begs for 20 minutes after being ignored shows higher resistance than a dog that stops after 5 minutes. Numerous factors influence this persistence.

The **most important factor** influencing resistance to extinction is the **schedule of reinforcement** that was in place before the extinction procedure began.





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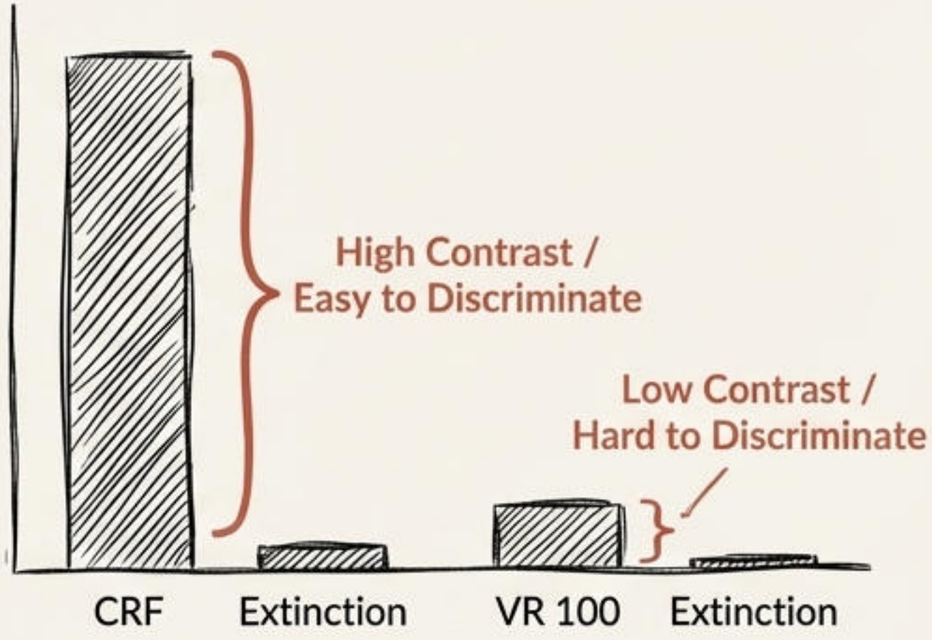
The Partial Reinforcement Effect (PRE)

The **partial reinforcement effect (PRE)** is the principle that behavior maintained on an **intermittent (partial) schedule** of reinforcement will **extinguish** more **slowly** than behavior maintained on a **continuous (CRF)** schedule.

- FR 100 behavior will persist longer than FR 10 behavior.
- CRF (or FR 1) behavior extinguishes the most rapidly.
- **Variable Ratio (VR) schedules** produce the highest resistance to extinction. This is why behaviors like gambling are so notoriously difficult to eliminate.

Why does the PRE occur?

- **Discrimination Hypothesis:** It is harder for the organism to "discover" that reinforcement is no longer available on an intermittent schedule. On a CRF schedule, the very first non-reinforced response is a clear signal that the contingency has changed. On a VR 100 schedule, long strings of non-reinforced responses are normal, so the onset of extinction is not immediately obvious.
- **Contrast Hypothesis:** There is a much greater contrast between CRF and extinction than between a lean intermittent schedule and extinction. The organism on a VR 100 schedule has a long history of emitting many responses in the absence of reinforcement, making it more persistent when extinction is implemented.



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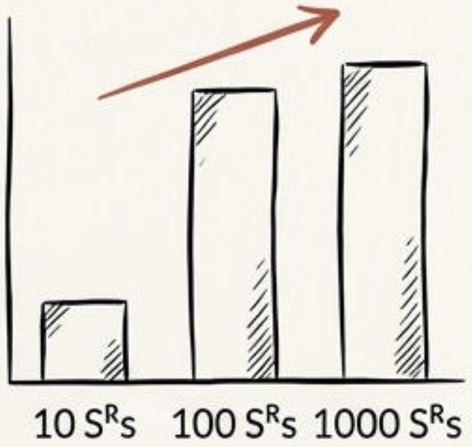
History and Magnitude of the Reinforcer

History of Reinforcement:

In general, the more reinforcers an individual has received for a behavior, the greater the resistance to extinction. A rat that has earned 100 food pellets will persist longer than a rat that has earned only 10.

- **Practical Implication:** It is far easier to extinguish a newly developed unwanted behavior than a long-standing one. This is the logic behind the adage, "nip it in the bud."
- **Caveat:** This effect is not infinite. Research with pigeons found that resistance to extinction reached its maximum after about 1,000 reinforcers (Furomoto, 1971).



Magnitude of the Reinforcer:

The effect of reinforcer size is less consistent.

- Sometimes, large-magnitude reinforcers (e.g., large food pellets, highly preferred food) lead to greater resistance to extinction.
- However, some studies have found the opposite, where smaller reinforcers produce greater resistance. The relationship is not entirely clear-cut.

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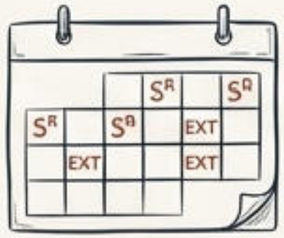
Deprivation, Previous Experience, and Signals

- **Degree of Deprivation:** The greater the level of deprivation for the reinforcer, the greater the resistance to extinction. A very hungry rat will press the lever longer than a slightly hungry rat.
  - **Practical Implication:** To extinguish a dog's begging, feeding the dog a full meal just before the family sits down to eat can reduce its motivation and thus its resistance.
- **Previous Experience with Extinction:** The more experience an organism has with extinction sessions being alternated with reinforcement sessions, the more quickly it will cease responding in future extinction sessions.
  - **Human Analogy:** Individuals who have been through numerous relationship breakups may learn to disengage more quickly and move on to new relationships, having learned the cues that signal a period of non-reinforcement.
- **Distinctive Signal for Extinction:** Extinction is facilitated when there is a clear **discriminative stimulus for extinction** (S^{Δ}). A signal that clearly indicates reinforcement is unavailable (e.g., a light turning off, a **"Closed" sign**) will cause responding to cease much more quickly.

Low Resistance



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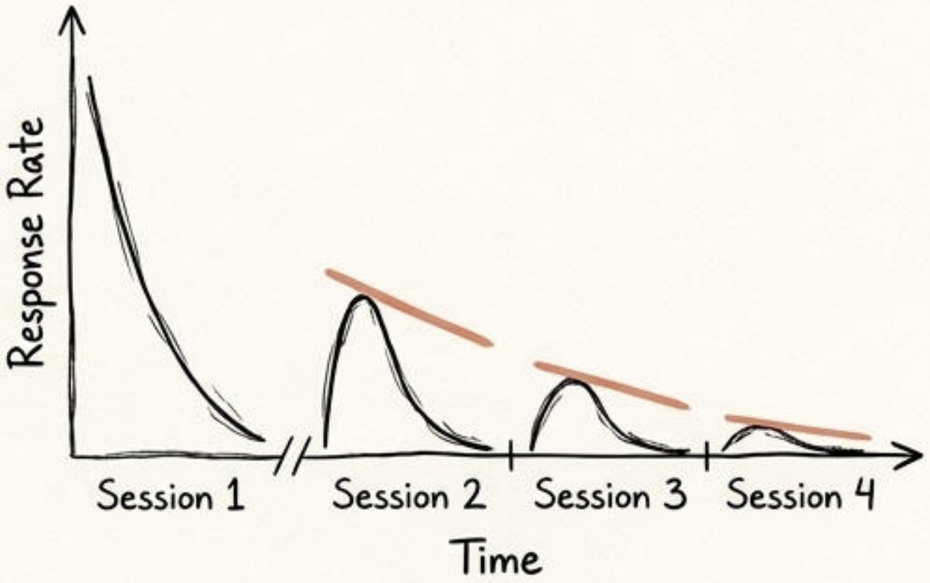
Recovery & Differential Reinforcement

Spontaneous Recovery

Just as in classical conditioning, an extinguished operant response is not permanently forgotten. **Spontaneous recovery** is the reappearance of an extinguished response following a rest period after extinction.

- **Process**: If a rat's lever pressing is extinguished in one session, placing it back in the chamber the next day will likely lead to a new bout of lever pressing.
- **Diminishing Returns**: This recovered response is typically weaker than before and will extinguish more quickly. Across successive extinction sessions, the amount of recovery becomes progressively smaller until the behavior is essentially eliminated.

B.F. Skinner (1950) proposed that this recovery is a function of **discriminative stimuli** (S^Ds) associated with the start of a session. For the rat, being taken from the cage, weighed, and placed in the chamber are all cues that have historically signaled the availability of reinforcement. Only after repeated exposure to these cues without reinforcement do they lose their function. For a child, simply entering the supermarket is an S^D for candy being available.



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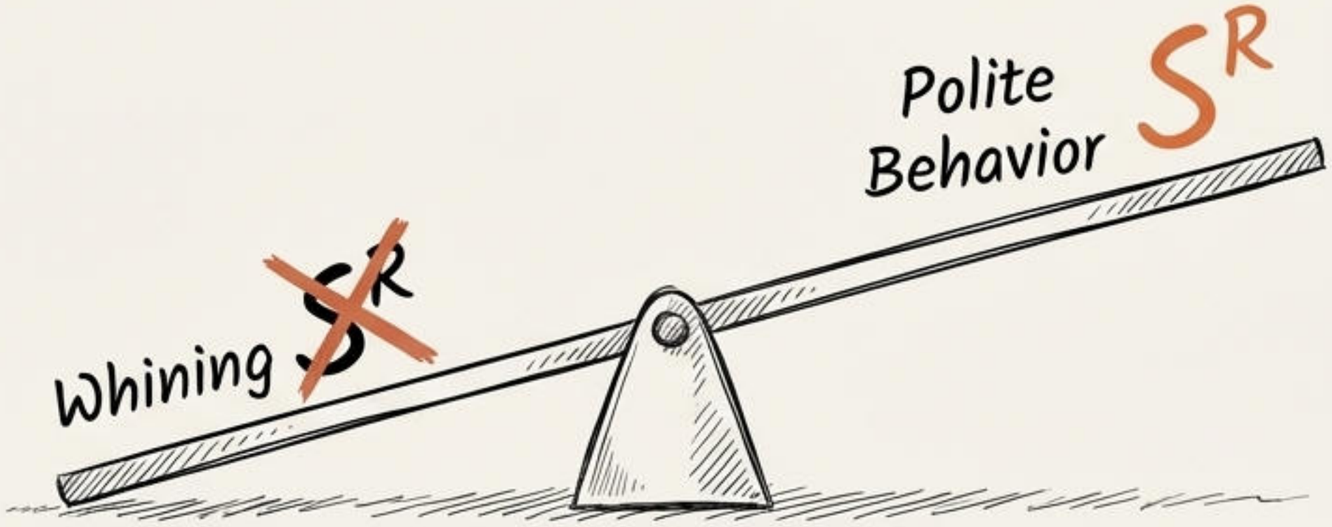
Differential Reinforcement of Other Behavior (DRO)

The process of extinction can be made significantly more effective and **less aversive** by pairing it with reinforcement for alternative behaviors. **DRO** is the procedure of reinforcing **any behavior other than** the target behavior that is being extinguished.

- **Mechanism:** DRO is highly effective because it attacks the problem from two directions: the target behavior is weakened by the lack of reinforcement, while alternative behaviors are strengthened by the presence of reinforcement, which come to replace the problem behavior.
- **Benefits:** A DRO procedure reduces or eliminates the challenging side effects of extinction (frustration, aggression, depression). This is because the individual is not being deprived of all reinforcement in that context, only for the specific unwanted behavior. For the child in the supermarket, you can extinguish whining while simultaneously providing the sought-after candy, but now contingent on displaying appropriate, well-mannered behavior. The candy can then be gradually faded out as polite behavior becomes established.



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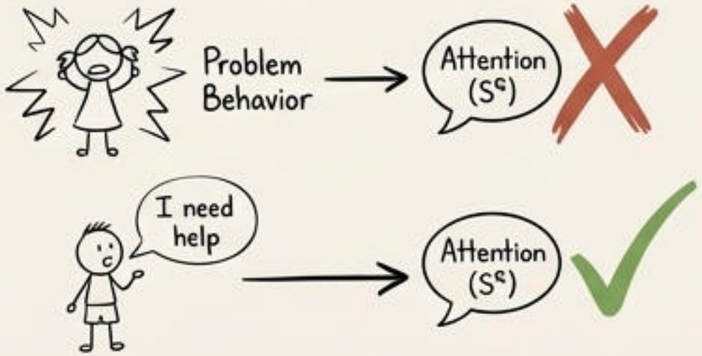
Functional Communication Training (FCT)

A particularly powerful and constructive form of differential reinforcement is **Functional Communication Training.** This approach is based on the understanding that many problem behaviors are simply inappropriate attempts to communicate a need and access a reinforcer (e.g., attention, escape from a task).

FCT involves teaching the individual a clear, socially appropriate communicative behavior to access that same reinforcer. The appropriate communication is then differentially reinforced, while the problem behavior is placed on extinction.

- **Example:** A child who misbehaves to get a parent's attention is taught to say, "I'm bored, can you help me find something to do?" When the child uses this phrase, the parent provides attention. When the child misbehaves, the behavior is ignored.

As a general clinical rule: whenever extinguishing an unwanted behavior, one should simultaneously and generously reinforce more appropriate behaviors. This principle of providing reinforcement independent of the problem behavior shares conceptual ground with the humanistic psychology concept of ***Unconditional Positive Regard*** (Rogers), which can be seen as a form of noncontingent reinforcement that supports well-being and reduces the need for problem behaviors.



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